Tutorial 5 – Collective Decision-Making: Urn Model, Global Switching, Foraging

Deadline: 12.07.2026

Objectives:

- Implement an empirical approach to the application of the urn model
- Test a simple model for how accurately it represents reality
- Implement and investigate foraging and measure swarm performance

1 Urn Model for Locust Scenario

We use the locust simulation discussed in the previous assignment (tutorial 4, task 1) again. This week we work with the following parameters: circumference $C = 0.5$, speed of 0.01, perception range of $r = 0.045$, swarm size of $N = 50$, spontaneous direction switch with a probability $P = 0.15$ per time step.

In your simulation you still have to keep track of the left-goers L_t at a time step t and count them. We are interested in how the average change in the number of left-moving locusts L_t depends on L_t itself. That is, we are searching for a function $\Delta L(L)$. To measure this function, we simulate the locusts for 100 time steps. These 100 time steps allow the locusts to settle into a stable collective pattern and organize themselves. Then, we simulate 20 further time steps while continuously tracking L_t. At each time step $t \in [101, 120]$ we store the one-step change in L, defined as $\Delta L = L_t - L_{t-1}$. To obtain the function $\Delta L(L)$, we group each measured one-step change ΔL according to the corresponding value of L_{t-1} and compute the average over all occurrences. This requires storing all ΔL values in bins indexed by L and keeping track of how many samples fall into each bin. To obtain reliable statistics, we repeat the simulation many times e.g. 5000 independent simulation runs (efficient code can reduce the runtime). After collecting sufficient samples, we compute the mean $\Delta L(L)$ for every possible L and save the resulting function for later analysis.

a) Plot the data for $\Delta L(L)$. What is the mathematical and domain-specific meaning of the positions L^* with $\Delta L(L^*) = 0$?

b) Next we fit the swarm urn model to this data. Remember the equations:

$$P_{FB}(s, \varphi) = \varphi \sin(\pi s), \tag{1}$$

$$\Delta s(s) = 4\left(P_{FB}(s, \varphi) - \frac{1}{2}\right)\left(s - \frac{1}{2}\right), \tag{2}$$

where P_{FB} is the probability of positive feedback and $\Delta s(s)$ is the expected average change of the considered state variable. Here these are the left-goers and hence s is the ratio of left-goers $s = L/N$. In addition, we have to introduce a scaling constant c to the idealized equation, which yields:

$$\Delta s(s) = 4c\left(P_{FB}(s, \varphi) - \frac{1}{2}\right)\left(s - \frac{1}{2}\right) \tag{3}$$

Fit this function to your data, that is, try to find appropriate values for the scaling constant c and the feedback intensity φ. Be sure to keep φ within a reasonable interval: $\varphi \in [0, 1]$.

c) Plot your data together with the fitted function. Also plot the resulting probability of positive feedback P_{FB}. What can be said about these results?

2 Density-Dependent Global Switching

We use the locust simulation once more. In the biological system with actual locusts a density-dependent global switching behavior was observed. Even though the swarm had converged to either a majority of left-goers or right-goers, it switched sometimes to the other majority. For low densities this was observed more frequently than for high densities. Here we test whether we see the same in our little simulation.

We use the parameter set as in task 1 except for the swarm size N that we want to vary. We test a number of swarm sizes on the interval $N \in [20, 150]$. We measure times between observed global switches in the following way. For a given swarm size N we define three 'zones' depending on the observed left-goers L:

Zone A: $L > 0.7N$

Zone B: $0.3N \leq L \leq 0.7N$

Zone C: $L < 0.3N$

We measure the times it takes the swarm to go either from zone A to zone C or from zone C to zone A. Whenever the swarm is in zone A or zone C we set a counter to zero. Once we enter zone B we start to count and remember from which zone we came from. When we then leave zone B into the other zone (i.e., the zone we did not come from), we store the time on the counter, set it to zero, and continue. If we leave zone B into the zone we came from we reset the counter to zero without storing the time.

a) Measure these durations for a number of swarm sizes $N \in [20, 150]$ for several independent simulation runs each, with each running for at least 5,000 time steps, but better more. Make sure you observe enough switches for each tested swarm size. Calculate the average observed global switch time for each tested swarm size and plot them over swarm size. Also plot the number of observed switches over swarm size. Do we observe something similar as in the natural swarm?

3 Foraging

In this task we go back to our collective robotics simulation. Implement a robot controller for foraging that satisfies the following specifications:

1. The robot is supposed to collect objects and take them to a home zone marked with bright light.
2. The robot has a number of proximity sensors and in addition a special kind of bumper sensor to detect when it is pushing against objects. A force is measured that allows to tell how many objects it is pushing (assuming all objects have the same size, weight, and friction etc.).
3. The robot has also two light sensors (left and right).
4. Your controller code that is executed in each cycle should output the following values:
   - i) Actuator value for motor left/right
   - ii) Error (false, true, or unknown)
   - iii) Collision (false, true, or unknown)
   - iv) Arena boundary (false, true, or unknown)
   - v) Transporting object (false, true, or unknown)
   - vi) Home zone (false, true, or unknown)

- Run the experiment with different swarm sizes $N \in \{1, 2, \ldots, 10\}$ (or even more). Measure the swarm performance (collected objects during a fixed period of time). Plot the swarm performance over swarm size.

Submission Requirements:

- Please zip your submission in a single file named:
  `collRob_sheet5_YOURLASTNAME1_YOURLASTNAME2.zip`
- Include the source code and a README explaining how to compile and run the applications
- You must provide a PDF report that includes the simulation details, analysis, conclusions, and relevant plots
- Optional but preferable: include a video where you go through and explain your code (but this is not a substitute for well-commented code or the mandatory report)